Beyond Pareto Fronts: A Leximax Universal-Regret Core for Robust Multicriteria Decision Support

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Abstract

Pareto optimality is a safety property, not a complete decision rule: an efficient set tells a decision maker (DM) which alternatives are not obviously bad, but offers no principled way to choose one. Practice fills this gap with *post hoc* devices—weighted sums, TOPSIS, compromise programming, knee selection, outranking thresholds, and hypervolume-based rules—each of which requires additional modelling choices such as weights, metrics, thresholds, exponents, or reference points. These choices are legitimate but may be hard to justify when preference information is intentionally incomplete. We propose the **Leximax Universal-Regret (LUR) core**, a declared-probe robust MCDA rule that ranks feasible alternatives by the lexicographically sorted vector of normalised disappointments over a finite family of monotone probes. LUR returns a single recommendation or a small tolerance-induced indifference class, together with a labelled regret certificate identifying the most binding criteria coalitions. We prove existence and completeness, Pareto compatibility *under a separation condition*, dominated-solution exclusion, stability of the tolerance-based leximax rule under bounded perturbations, and a probe perturbation bound with explicit constants, and we give an adaptive correlation-clustering construction empirically reducing, without eliminating, sensitivity to redundant criteria. In a replicated benchmark and a pre-registered broadened protocol covering 11 methods, 10 held-out preference families, 8 front geometries, and m up to 20, LUR is practically equivalent to the best robust scalarisations on tail held-out loss, while remaining clearly ahead of distance-based and Monte-Carlo single-choice baselines. Finally, we allow stochastic and multi-stakeholder variants, which we treat as initial extensions rather than complete standalone theories. All code, data and scripts that regenerate every table and figure are released.

Keywords: Decision support; Multicriteria decision analysis; Robust decision making; Minimax regret; Achievement scalarising functions; Pareto efficiency.

1 Introduction

1.1 The Pareto front as a safety filter, not a decision object

Since the work of Pareto, efficiency has been the cornerstone of multicriteria analysis: an alternative is Pareto efficient if no criterion can be improved without worsening another. This is a logically impeccable *safety property*—it removes alternatives that are dominated and therefore indefensible. It is, however, *decisionally incomplete*. For m conflicting criteria the efficient set is typically large or infinite, and the decision maker (DM) is left with a cloud of efficient alternatives and no principled rule for selecting one.

Confronted with this gap, practice resorts to *post hoc* selection devices, each of which reintroduces the very preference information the DM was unable to supply: a weighted sum requires weights and cannot reach non-convex parts of the front (Miettinen, 1999); TOPSIS (Hwang and Yoon, 1981) requires a distance metric and weights; compromise programming (Zeleny, 1982) requires a metric exponent and weights; knee-point rules (Branke et al., 2008) are scale sensitive; outranking methods such as ELECTRE (Roy, 1991) and PROMETHEE (Brans and De Smet, 2016) require concordance/discordance thresholds; and hypervolume-based selection (Zitzler and Thiele, 1999) favours extremes and is costly. These methods are valid and widely used, but they require the analyst or DM to commit to additional modelling choices—weights, metrics, thresholds, exponents, or reference points—that may be difficult to justify when preference information is intentionally incomplete. The fundamental issue is that **the Pareto front is a safety filter, not a complete decision rule**.

1.2 From a front to a certificate

We argue that, for the purpose of *choosing*, the efficient set is often insufficient as a final decision object. We propose supplementing it with a *regret certificate*: a vector of normalised disappointments over the rational questions the DM might reasonably ask about the criteria, sorted from worst to best. Rather than asking "which alternatives are efficient?" we ask

> *which alternative is least fragile across all declared rational interpretations of the criteria?*

This turns the output of the analysis from a *set* into a *certified recommendation*: a single alternative (or small indifference class) accompanied by an explanation of which questions are most disappointed and by how much.

It is crucial to note what this approach does and does not do. LUR is not parameter-free. It replaces a single preference vector by an explicit family of admissible probes, and it requires normalisation bounds and tolerance settings for numerical implementation. Its contribution is that these choices are declared, auditable, and separated from the final recommendation, not that they disappear.

1.3 The Leximax Universal-Regret core

We formalise this as the **Leximax Universal-Regret (LUR) core**. Given a feasible set and a *declared* class $\mathcal{Q}$ of admissible monotone probes (scalar "rational questions" over the normalised criteria), LUR ranks alternatives by the lexicographically sorted vector of their normalised regrets across $\mathcal{Q}$, and returns the lexicographic minimiser together with its certificate. The construction rests on three principles: (i) *declared admissibility*—$\mathcal{Q}$ is part of the problem statement, not hidden inside a scalarisation; (ii) *Pareto compatibility*—under monotone probes a dominated alternative is never preferred to its dominator; and (iii) *direct finite computation*—the recommendation is obtained exactly on a finite candidate set without further *post hoc* preference search; for structured continuous cases, §4 gives conditional front-free formulations whose large-scale validation remains future work.

1.4 Positioning and contribution

LUR belongs to *robust* multicriteria decision analysis. Its closest relatives are stochastic multicriteria acceptability analysis (SMAA) (Lahdelma et al., 1998; Tervonen and Figueira, 2007), which integrates acceptability over a preference distribution; robust ordinal regression (Greco et al., 2008; Figueira et al., 2009), which reasons over the set of value functions compatible with elicited holistic judgements; and preference robust optimisation (Armbruster and Delage, 2015; Hu and Mehrotra, 2015), which optimises against an ambiguity set of utility functions. LUR differs from all three in object and output: it computes the *lexicographic minimax of normalised regret* over an explicit, monotone probe family and returns an interpretable certificate; it requires no preference distribution, no elicited judgements, and no single scalarisation. We show that achievement scalarisation (Wierzbicki, 1980), the weighted sum, and the lexicographic maximin (Ogryczak, 1997) are recovered as special cases of the probe family, so LUR is a *unifying* as well as a new rule.

Our contributions are:

- **Core theoretical contribution:** We define the LUR order and its auditable certificate, proving existence, completeness, Pareto compatibility *under an explicit separation condition*, dominated-solution exclusion, and stability of the tolerance-based rule under bounded perturbations (§3, Appendix A).
- **Computational contribution:** We present an exact candidate-set algorithm based on sequential lexicographic sorts, and outline a conditional formulation for structured continuous cases (§4).
- **Probe construction:** We propose an adaptive, redundancy-aware correlation-clustering probe family with empirical redundancy tests and a perturbation bound (§5).
- **Empirical validation:** We validate LUR on a broadened benchmark suite with held-out additive and non-additive preferences, and apply it to a stochastic smart-grid dispatch case study (§7).
- **Extensions:** We outline stochastic and multi-stakeholder variants as initial extensions demonstrating the framework's flexibility (§6).

The remainder of the paper reviews related work (§2), develops the core and its guarantees (§3), the direct formulation (§4), the adaptive probes (§5) and the extensions (§6), reports the computational study (§7), discusses limitations (§8) and concludes (§9). All proofs are in Appendix A.

2 Background and related work

2.1 Pareto efficiency and *post hoc* selection

For a vector objective $F(x) = (f_1(x), \ldots, f_m(x))$ (minimisation), x Pareto-dominates y if $f_i(x) \leq f_i(y)$ for all i with at least one strict inequality; the efficient set is the non-dominated set. Because it is typically large, a single recommendation is obtained by *post hoc* selection. The weighted sum, TOPSIS (Hwang and Yoon, 1981), compromise programming (Zeleny, 1982), knee selection (Branke et al., 2008), ELECTRE (Roy, 1991), PROMETHEE (Brans and De Smet, 2016) and hypervolume selection (Zitzler and Thiele, 1999) are the standard tools; each requires weights, thresholds, a metric, or an already generated front. LUR does not eliminate modelling choices. It relocates them into a declared family of admissible monotone probes and, on finite candidate sets, computes the resulting recommendation exactly while returning a labelled regret certificate.

2.2 Achievement scalarising functions and ordered aggregation

Achievement scalarising functions (ASF) project a reference point onto the efficient set through a parameterised metric interpolating between L_1 and L_∞ (Wierzbicki, 1980; Miettinen and Mäkelä, 2002). Ordered weighted averaging (Yager, 1988) aggregates sorted criteria, and Ogryczak (1997); Ogryczak and Śliwiński (2006) connect ordered aggregation to lexicographic maximin. LUR *recovers* these as special cases when the probe family is reduced to a single probe: the grand-maximum probe alone gives the (reference-at-ideal) Chebyshev/ASF rule, the grand-mean probe alone the weighted sum, and the singleton family the lexicographic maximin of normalised regret. We do not claim LUR subsumes these methods with its full adaptive family—there it is a *different* aggregation paradigm. Its distinguishing features are that it sorts *normalised regrets* (dimensionless disappointments relative to each probe's own attainable best and worst) over a *declared family of probes* and breaks ties through the whole sorted vector, and that it emits a certificate.

2.3 Robust MCDA: SMAA, robust ordinal regression, preference robust optimisation

The work closest to LUR lies in *robust* multicriteria decision analysis. **SMAA** (Lahdelma et al., 1998; Tervonen and Figueira, 2007) computes, by Monte-Carlo integration over a distribution of admissible weights (and possibly criterion values), acceptability indices describing how strongly each alternative is supported. **Robust ordinal regression** (Greco et al., 2008; Figueira et al., 2009) reasons over the entire set of additive value functions compatible with a DM's holistic pairwise judgements, producing necessary and possible preference relations. **Preference robust optimisation** (Armbruster and Delage, 2015; Hu and Mehrotra, 2015) optimises the worst-case expected utility over an ambiguity set of utility functions. LUR shares the robust philosophy—perform well across a whole class of admissible preferences rather than commit to one—but differs in three ways. First, its admissible class is an explicit family of *monotone probes*, not a weight distribution or an elicited polyhedron. Second, its criterion is the *lexicographic minimax* *of normalised regret*, not an expected acceptability, a necessary/possible relation, or a single worst-case utility. Third, its output is an interpretable *certificate* naming the binding coalitions. We note that SMAA is primarily an *exploratory* framework that informs the DM through acceptability indices rather than emitting a single choice; to obtain a comparable single-choice rule we use the standard SMAA-2 device of recommending the alternative with the highest first-rank acceptability index (ties broken by expected value), and we treat the comparison as one of decision philosophies as much as a benchmark. We likewise compare against a Savage minimax-regret (MMR) recommender over a sampled linear weight set (§7). More broadly, LUR is complementary to data-driven preference disaggregation and inverse MCDA (Doumpos and Zopounidis, 2011), which infer preference parameters from holistic judgements: LUR instead dispenses with a fitted preference model and reasons over a declared class of questions, and could take an elicited probe family as input where such data exist.

2.4 Minimax regret and robust optimisation

Regret-based choice originates with Savage (1951) and Loomes and Sugden (1982); minimax regret seeks alternatives minimising the largest shortfall from the best attainable value. Robust optimisation (Ben-Tal and Nemirovski, 2002) and stochastic programming (Birge and Louveaux, 2011) address parameter uncertainty in the model rather than preference uncertainty in the DM. LUR extends minimax regret in three directions: it is *lexicographic* (after the worst regret, minimise the second worst, and so on), *probe-based* (regret is computed over rational coalitions of criteria, not raw criteria), and *direct* (computed without enumerating the efficient set).

2.5 Social choice and fairness

Aggregating criteria resembles social choice, but treating criteria as voters invites Condorcet cycles; LUR avoids this by aggregating *regrets over declared questions*, not votes over criteria. For multi-stakeholder settings we draw on the maximin principle of Rawls (1971) and the bargaining viewpoint of Nash (1950); we claim natural fairness properties (anonymity, monotonicity, worst-stakeholder protection), not a uniqueness theorem.

2.6 What is new relative to existing regret and robust-MCDA rules

While LUR shares motivation with several robust approaches, its mathematical object and output differ fundamentally:

- **Versus minimax regret:** LUR minimises a sorted vector of normalised probe regrets, not a single maximum regret over raw utilities or sampled weights.
- **Versus ASF/Chebyshev:** ASF commits to one scalarising function or one reference structure; LUR evaluates a declared family and certifies binding probes.
- **Versus SMAA:** SMAA integrates over a preference distribution and reports acceptability; LUR does not require a probability distribution and returns a leximax regret certificate.

Table 1: Comparison of LUR with representative selection and robust-MCDA methods.

Method	Main preference / model input	Requires candidate/front?	Single rec.?	Pareto comp.	Explanation / certificate	Main limitation
Weighted sum	Weight vector	No	Yes	Yes ($w > 0$)	Low	Cannot recover non-supported points in non-convex fronts
TOPSIS	Weights and distance metric	Yes	Yes	Not strictly	Low	Rank reversal; distance to non-feasible ideal
Compromise programming	Weights and L_p norm exponent	Yes	Yes	Yes ($p < \infty$)	Low	Non-linear scaling effects
ASF / Chebyshev	Reference point and scaling	No	Yes	Yes	Low	Requires setting an arbitrary reference point
MMR (sampled weights)	Linear weight distribution	Yes	Yes	Yes	Worst-case regret value	Limited to linear aggregations
SMAA	Distribution over preferences/weights	Usually	Not inherently (needs rule)	Usually	Acceptability indices	Exploratory; single-choice rule depends on post-processing
VIKOR	Weights and majority threshold	Yes	Yes	Not strictly	Low	Complex parameter interaction
Hypervolume contrib.	Reference point	Yes	Yes	Yes	Low	Favours extremes; sensitive to reference point
LUR (ours)	Declared monotone probe family; bounds; tolerance τ; optional θ	Candidate-set exact; continuous cond.	Yes (or small τ-class)	Yes (if separating probes)	Full labelled regret certificate	Sensitive to probe family, normalisation, and near-ties

- **Versus robust ordinal regression (ROR):** ROR reasons over compatible value functions after elicited comparisons; LUR can operate without holistic preference judgements.
- **Versus preference robust optimisation (PRO):** PRO optimises worst-case utility over an ambiguity set; LUR optimises lexicographically sorted normalised disappointments over interpretable monotone probes.
- **Versus ordered weighted aggregation (OWA):** LUR sorts regrets across questions, not criterion values themselves.

The novelty is not minimax regret alone, but lexicographic minimax over normalised disappointments induced by a declared monotone probe family, with an auditable certificate and an adaptive redundancy-aware probe construction.

Table 1 summarises where LUR sits relative to representative MCDA and robust methods.

3 The Leximax Universal-Regret core

3.1 Problem setting and normalisation

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be the feasible set and $F : \mathcal{X} \to \mathbb{R}^m$, $F(x) = (f_1(x), \ldots, f_m(x))$, the vector objective.

Convention: All criteria f_i are assumed to be *minimised*. Maximisation objectives must have their sign reversed before normalisation. Consequently, lower normalised values r_i are better, and probes are non-decreasing functions of these costs.

For the finite case let $A = \{x_1, \dots, x_N\} \subseteq \mathcal{X}$ be a candidate set, e.g. an *a posteriori* approximation of the efficient set; for the continuous case (§4) A is not required. Criteria are normalised to $[0,1]$ by the positive affine map

$$r_i(x) = \frac{f_i(x) - z_i^\star}{\max\{z_i^N - z_i^\star,\ \varepsilon\}}, \qquad z_i^\star = \min_{x \in A} f_i(x), \ \ z_i^N = \max_{x \in A} f_i(x), \tag{1}$$

attains both endpoints on A. If $z_i^N - z_i^\star$ approaches zero, the normalisation floor ε prevents division by zero and effectively zeros out the relevance of that criterion. A criterion is flagged as numerically degenerate when $z_i^N - z_i^\star < \kappa \max\{1, |z_i^\star|, |z_i^N|\}$, with κ set to a small numerical tolerance such as 10^{-8}. Degenerate criteria are reported and excluded from non-singleton probe construction unless the analyst supplies external bounds.

For the finite case all extrema ($z^\star$, z^N, $q^\star$, q^{w} below) are taken over the candidate set A; for the continuous case (§4) they are taken over $\mathcal{X}$. We use the worst attainable value over A in place of the true nadir over the efficient set, the two coincide when A contains the criterion-wise worst efficient points and otherwise the former is a conservative bound. The rule is invariant to positive affine rescaling of the criteria when the bounds are transformed consistently; it is not invariant to arbitrary monotone transformations (e.g. $f_i \mapsto \log f_i$), which change the distances that define regret—so such transformations are a modelling choice the analyst must make deliberately.

3.2 Admissible probes

A *probe* is a scalar function $q : [0,1]^m \to \mathbb{R}$ encoding a rational decision question. The family $\mathcal{Q}$ is *declared*: it is part of the specification.

Definition 1 (Admissible probe)**.** *A probe q is* admissible *if (i) it is non-decreasing in each r_i (monotonicity); (ii) it can be rescaled so that $q(\mathbf{0}) = 0$, $q(\mathbf{1}) = 1$ (normalisation); and (iii) it is L-Lipschitz in $\|\cdot\|_\infty$.*

Here $\mathbf{0}$ and $\mathbf{1}$ denote the all-zeros and all-ones vectors. Monotonicity gives *weak* Pareto compatibility; strict compatibility needs a separation property (Definition 2). Canonical admissible probes are the *singletons* $q_i(r) = r_i$; the *grand mean* $q_\mu(r) = \frac{1}{m}\sum_i r_i$ (the utilitarian question); the *grand maximum* $q_{\max}(r) = \max_i r_i$ (the egalitarian/worst-criterion question); and, for any coalition $S \subseteq \{1, \dots, m\}$, the coalition mean and maximum. All have non-negative weights or are maxima, hence are monotone.

Definition 2 (Separating family)**.** *A probe family $\mathcal{Q}$* separates *criteria if for every i there is a probe $q \in \mathcal{Q}$ strictly increasing in r_i. Including all m singleton probes $q_i(r) = r_i$ is sufficient; LUR always does so.*

3.3 Disappointment and regret

For each probe $q \in \mathcal{Q}$ let $q^\star = \min_x q(r(x))$ and $q^{\mathrm{w}} = \max_x q(r(x))$ (or a user-supplied worst-acceptable value). The *normalised disappointment* of x under q is

$$D_q(x) = \frac{q(r(x)) - q^\star}{q^{\mathrm{w}} - q^\star + \varepsilon} \in [0,1], \tag{2}$$

with $D_q(x) = 0$ iff x is q-optimal and 1 iff x is q-worst.

We use the term *disappointment* for the normalised shortfall of an alternative from the best attainable value of a single given probe, and *regret certificate* for the vector collecting these probe-wise disappointments. This is regret-like but differs from classical pairwise Savage regret unless the probe family is interpreted specifically as the uncertainty set of possible true preference functions.

3.4 The leximax order

Collect $D(x) = (D_{q_1}(x), \ldots, D_{q_K}(x))$, $K = |\mathcal{Q}|$, and sort it in *descending* order to obtain $D^{\downarrow}(x) = (D_{[1]}(x) \geq \cdots \geq D_{[K]}(x))$. The **Leximax Universal-Regret order** is

$$x \preceq_{\mathrm{LUR}} y \iff D^{\downarrow}(x) \leq_{\mathrm{lex}} D^{\downarrow}(y), \tag{3}$$

i.e. x is preferred if its worst disappointment is smaller; on ties, its second worst; and so on. The *LUR solution* is $x^{\mathrm{LUR}} \in \arg\min_x^{\mathrm{lex}} D^{\downarrow}(x)$, and its *certificate* is the labelled, sorted vector $D^{\downarrow}(x^{\mathrm{LUR}})$ together with the probes attaining the largest entries.

Because exact lexicographic comparison is discontinuous at ties, we use in practice—and in all guarantees below—a *tolerance-based* leximax order: fix $\tau > 0$ and compare $D^{\downarrow}(x)$ and $D^{\downarrow}(y)$ at the first coordinate where they differ by more than τ, treating coordinates within τ as equal. This is the rule implemented in the released code and the one for which stability holds (Theorem 3).

3.5 Foundational guarantees

Theorem 1 (Existence and completeness)**.** *For any finite A and finite $\mathcal{Q}$, x^{LUR} exists and $\preceq_{\mathrm{LUR}}$ is a complete preorder on A.*

Theorem 2 (Weak and strict Pareto compatibility)**.** *Let x strictly Pareto-dominate y and let every $q \in \mathcal{Q}$ be monotone non-decreasing in each r_i. Then $D_q(x) \leq D_q(y)$ for all q (*weak *compatibility), so $x \preceq_{\mathrm{LUR}} y$. If, in addition, $\mathcal{Q}$* separates *criteria (Definition 2)—in particular if $\mathcal{Q}$ contains the singleton probes—then $D_q(x) < D_q(y)$ for some q and $x \prec_{\mathrm{LUR}} y$.*

The separation hypothesis is necessary: with only the probe $q(r) = r_2$, the dominator $r(x) = (0, 1)$ and the dominated $r(y) = (0.1, 1)$ have equal disappointment, so monotonicity alone does not give a strict preference. LUR always includes the singletons, so it separates.

Corollary 1 (Dominated exclusion)**.** *If x is Pareto-dominated by a feasible y and $\mathcal{Q}$ separates criteria, then x is not LUR-optimal.*

Theorem 3 (Stability of the tolerance-based rule)**.** *Use the τ-tolerance leximax order. Let $\hat{D}_q(x)$ estimate $D_q(x)$ with $\eta = \max_{x,q} |\hat{D}_q(x) - D_q(x)|$. Suppose the winner x^{LUR} beats every rival y at the first coordinate $k(y)$ at which their sorted vectors differ by more than τ, with margin $D_{[k(y)]}(y) - D_{[k(y)]}(x^{\mathrm{LUR}}) \geq \Delta$, and that all earlier coordinates agree within $\tau - 2\eta$. If $2\eta < \min\{\Delta - \tau,\ \tau\}$, the winner under $\hat{D}$ is unchanged.*

Theorem 3 is deliberately stated for the tolerance rule: exact lexicographic comparison is *not* perturbation-stable, because a winner that beats a rival only at a later coordinate after an exact earlier tie can be reversed by an arbitrarily small perturbation of the earlier coordinate. The tolerance τ absorbs exactly those near-ties. Proofs are in Appendix A. For continuous $\mathcal{X}$, existence requires compactness of $\mathcal{X}$ and continuity of each $q \circ r$; uniqueness is not guaranteed and a small indifference class may arise.

4 Exact candidate-set computation and conditional direct formulations

LUR is primarily a candidate-set selection rule, and that is the setting of our experiments. We first present the exact finite algorithm. We then develop, more tentatively, how the recommendation can in principle be computed *directly* on a continuous feasible set, replacing the usual two-stage pipeline by a single robust search. We present the continuous case as a formulation with assumptions whose large-scale empirical validation we leave to future work.

4.1 Exact algorithm for finite candidate sets

Given a discrete candidate set $A = \{x_1, \ldots, x_N\}$: (1) form $F_{ij} = f_j(x_i)$; (2) normalise to R via (1); (3) evaluate $V_{iq} = q(r(x_i))$ for each probe; (4) set $q^\star = \min_i V_{iq}$, $q^{\mathrm{w}} = \max_i V_{iq}$; (5) form disappointments (2); (6) sort each row descending; (7) return the lexicographically smallest row.

The worst-case complexity is straightforward. With naive probe evaluation, winner selection costs $O(NKm+NK\log K+NK)$. Producing a complete lexicographic ranking of all candidates costs $O(NKm + NK\log K + NK\log N)$. Here $O(NKm)$ is the cost of probe evaluation and $O(NK\log K)$ is the cost of sorting the K-dimensional disappointment vector for each of the N candidates. The number of probes K is controlled by the construction of §5.

4.2 Cost of the anchors

Both the finite and continuous formulations need, for each probe, its attainable best $q^\star = \min_x q(r(x))$ and worst $q^{\mathrm{w}} = \max_x q(r(x))$. On a candidate set these are $O(N)$ scans; on a continuous $\mathcal{X}$ each is a scalar global optimisation, so the family contributes $2K$ such subproblems—cheap relative to front enumeration when K is linear in m (§5) but not free, and we account for it explicitly.

4.3 Exact sequential cumulative-sum formulation

Lexicographic minimisation of the sorted vector is exactly equivalent to sequentially minimising the cumulative sums of the p largest disappointments. For each $p = 1, \ldots, K$, define:

$$S_p(x) = \sum_{k=1}^{p} D_{[k]}(x), \tag{4}$$

where $D_{[1]}(x) \geq \cdots \geq D_{[K]}(x)$ are the sorted disappointments. Then solve sequentially:

$$\rho_p = \min_{x \in \mathcal{X}_{p-1}} S_p(x), \quad \mathcal{X}_p = \{x \in \mathcal{X}_{p-1} : S_p(x) = \rho_p\}, \tag{5}$$

starting with $\mathcal{X}_0 = \mathcal{X}$. Each cumulative sum is convex when the original disappointments are convex and admits the standard ordered-value linearisation (Ogryczak and Śliwiński, 2006). By solving K sequential problems, we obtain the exact LUR recommendation without needing to enumerate the Pareto front.

4.4 Single-stage heuristic surrogate

The stages can be collapsed with a strictly decreasing weight schedule. For a large constant $M > 1$,

$$\min_{x \in \mathcal{X},\, \rho_1 \geq \cdots \geq \rho_K \geq 0} \sum_{k=1}^{K} M^{-(k-1)} \rho_k \quad \text{s.t.} \quad D_{(k)}(x) \leq \rho_k, \tag{6}$$

where $D_{(k)}(x)$ denotes the k-th largest disappointment at x. This is an *exact* surrogate for the lexicographic order only under finite-gap assumptions. For general continuous values no finite M reproduces the exact order, so this single-stage form is primarily a heuristic approximation.

4.5 Smoothed continuous optimisation

For continuous $\mathcal{X} \subseteq \mathbb{R}^n$ the inner maximum is non-smooth; a log-sum-exp surrogate

$$\max_{q \in \mathcal{Q}} D_q(x) \approx \tfrac{1}{\tau} \log\Big(\textstyle\sum_{q \in \mathcal{Q}} \exp\big(\tau\, D_q(x)\big)\Big) \tag{7}$$

is smooth and approaches the maximum as $\tau \to \infty$. A probe-active-set sequential convex program—linearise the active probes $\mathcal{Q}_k = \{q : D_q(x_k) \geq \rho_1 - \varepsilon\}$ at the current iterate, solve the resulting convex subproblem, update, and repeat—then converges to a local LUR solution under standard convexity assumptions, analogously to SQP.

4.6 Why this matters, and what we do not yet show

The DM specifies the probe family once; in the finite case LUR returns the least-fragile compromise by the exact procedure above, and in the continuous case the formulations here make a front-free computation *available* in principle. We are careful not to overclaim: our experiments operate on candidate sets, so the large-scale continuous advantage is argued, not yet demonstrated, and we flag its We quantify the probe-family savings in §5. The direct formulations are empirically exercised in §7 on one continuous linear multiobjective problem and one mixed-integer facility-location problem. These tests demonstrate feasibility for structured cases, but not yet a general large-scale continuous optimisation advantage.

5 Adaptive, redundancy-aware probe generation

5.1 The redundancy problem

The full coalition family (mean and maximum over every non-trivial subset of criteria) has size exponential in m. Worse, real criteria are often *redundant*: several measured objectives track the same underlying concern and are strongly positively correlated. Averaging selection rules implicitly *over-weight* such a group—it is counted once per member—biasing the recommendation.

Consider a 3-criteria numerical example where f_1 is a weak independent criterion and f_2, f_3 are redundant metrics (e.g. both measure environmental impact):

Alternative	r_1	r_2	r_3	Interpretation
A	0.20	0.21	0.70	good redundant group, weak independent criterion
B	0.35	0.36	0.45	balanced
C	0.55	0.56	0.25	weak redundant group, good independent criterion

A full coalition family overcounts $\{r_1, r_2\}$ (which are highly correlated), implicitly double-weighting the environmental impact. The adaptive clustered family identifies the correlation and reduces overcounting by adding one cluster mean/max instead of many redundant coalitions. However, singleton probes remain (for separation), so perfect duplication invariance is not claimed; singleton influence usually appears in lower-order tie-breaking unless top regrets are tied. We need a probe family that (i) is small, (ii) stays monotone, and (iii) is less sensitive to redundancy than averaging aggregation.

5.2 Correlation clustering

We group *redundant* criteria by single-linkage connected components of the graph whose edges join pairs with Pearson correlation $\mathrm{Corr}(f_i, f_j) \geq \theta$. We use single-linkage because it treats redundancy as connected substitutability and avoids splitting a continuous underlying dimension into several independent concerns. This choice can, however, over-merge criteria through chaining; therefore we report sensitivity to θ and, in the supplementary material, compare with complete-linkage clustering.

Redundancy is positive correlation: two criteria that rise and fall together carry duplicate information, whereas negatively correlated criteria are genuine trade-offs and must remain separate—hence we threshold the signed correlation, not its absolute value. This yields clusters $C_1, \ldots, C_c$.

5.3 The adaptive probe family

LUR always includes the canonical anchors—the m singletons, the grand mean, and the grand maximum—and adds one mean and one maximum probe *per cluster* of size > 1:

$$\mathcal{Q} = \{q_i\}_{i=1}^{m} \cup \{q_\mu, q_{\max}\} \cup \bigcup_{|C_k|>1} \{q_{C_k,\mathrm{mean}}, q_{C_k,\max}\}. \tag{8}$$

A redundant group thus contributes *one* aggregate (cluster) question rather than a separate aggregate per member, so the family does not amplify a correlated group the way an averaging aggregate does, while the count stays $O(m+c)$—linear, not exponential. We stress that this *reduces*, but does not eliminate, redundancy sensitivity: the singleton probes for the group's members remain in $\mathcal{Q}$ (they are needed for separation, Theorem 2), so a redundant group can still influence lower-order lexicographic tie-breaking. The claim is therefore empirical—less redundancy bias than averaging methods on the tested benchmarks (§7)—not a duplication-invariance theorem. All probes are non-negative-weighted sums or maxima, hence monotone, so Pareto compatibility (Theorem 2) is preserved.

Theorem 4 (Non-negative-probe Pareto compatibility). *If every probe in $\mathcal{Q}$ is a non-negative-weighted sum or a maximum of the normalised criteria, then LUR is Pareto compatible.*

5.4 A perturbation bound (not a ranking guarantee)

We state the reduction effect as a regret-value perturbation lemma with explicit constants. We are deliberately modest: it bounds the change in individual disappointments, *not* the change in the LUR ranking or winner, and it depends on the normalisation floor ε, which is decision-critical.

Lemma 1 (Disappointment perturbation). *Suppose for a full-family probe q there is a reduced-family probe q' with $\sup_r |q(r) - q'(r)| \le \eta$ (hence also $|q^\star - q'^\star| \le \eta$ and $|q^{\mathrm{w}} - q'^{\mathrm{w}}| \le \eta$), and let the normalisation ranges satisfy $q^{\mathrm{w}} - q^\star,\ q'^{\mathrm{w}} - q'^\star \ge \beta > 0$. Then for every x*

$$|D_q(x) - D_{q'}(x)| \ \le\ \frac{3\eta}{\beta},$$

with the floor $\beta \ge \varepsilon$ in the worst case. The bound is vacuous when β is of order η, and informative only when the probe's attainable range is bounded away from zero—a regime the analyst can check directly.

Whether the reduced family admits such an η-close q' for every full-family probe is *not* established by correlation clustering in general; we therefore present Lemma 1 as a sensitivity statement and rely on the empirical redundancy study (§7) for evidence that clustering retains decision quality.

5.5 Empirical reduction

Table 2 reports the realised reduction on concave instances. The full family here is $\{q_{\mathrm{mean},S}, q_{\mathrm{max},S} : \emptyset \ne S \subseteq \{1,\dots,m\}\}$, of size $2(2^m - 1) - m$ (mean and max coincide on singletons); the adaptive family is linear in m, giving a nearly four-thousand -fold reduction by $m = 15$ with, as §7 shows, little loss of decision quality.

5.6 Greedy probe selection (optional)

When even the adaptive family is larger than needed, a greedy set-cover heuristic selects a minimal subfamily distinguishing all dominant pairs; for the max-coverage variant it attains a $(1 - 1/e)$ guarantee and for full coverage a $1 + \ln d$ guarantee (Nemhauser et al., 1978; Chvátal, 1979). We do not use it in the experiments, which already run on the compact adaptive family.

Table 2: Probe-family size: full coalition family ($2(2^m - 1) - m$ probes) vs. adaptive correlation-clustering family (concave instances, $N = 400$).

m	full coalitions	adaptive probes	reduction
3	11	5	2.2×
5	57	7	8.1×
8	502	10	50.2×
10	2,036	12	169.7×
15	65,519	17	3,854.1×

6 Extensions: stochastic and multi-stakeholder variants

6.1 Stochastic regret certificates

When criteria are noisy, model $f_i(x)$ as drawn from a joint distribution $P(x)$. Each probe value is then a random variable; for non-linear probes (e.g. $q_{\max}$) its mean $\mu_q(x)$ is *not* a simple function of the criterion marginal means. We estimate the robust stochastic probe value by its finite sample average over S scenarios $\omega^{(1)}, \dots, \omega^{(S)}$:

$$\hat{\mu}_q(x) = \frac{1}{S} \sum_{s=1}^{S} q(r(x, \omega^{(s)})). \tag{9}$$

The normalisation anchors $q^\star, q^{\mathrm{w}}$ are taken on these estimated means. We define the α-*confidence regret*

$$D_q^\alpha(x) = \frac{\mathrm{CVaR}_\alpha(q(r(x, \omega))) - q^\star}{q^{\mathrm{w}} - q^\star + \varepsilon}, \tag{10}$$

where CVaR_α is the empirical Conditional Value-at-Risk at confidence level α. This penalises both expected disappointment and its sample uncertainty. Because CVaR is a monotone risk measure, exact Pareto compatibility is preserved.

Proposition 1 (Scenario-wise Pareto compatibility)**.** *For finite A, finite $\mathcal{Q}$, and sample-average probe means $\hat{\mu}_q(x) = S^{-1} \sum_s q(r(x, \omega^{(s)}))$, if x weakly dominates y in every scenario and strictly dominates it in at least one criterion for at least one scenario, then the sample-average LUR rule does not prefer y to x. If $\mathcal{Q}$ separates criteria and the dominance is strict on a separated coordinate, then x is strictly preferred by the exact LUR order.*

Proposition 2 (Sample-average consistency under a margin condition)**.** *Let A and $\mathcal{Q}$ be finite. For each $x \in A$ and $q \in \mathcal{Q}$, suppose $q(r(x, \omega))$ is i.i.d. across scenarios and has finite variance. Let $\mu_q(x) = \mathbb{E}[q(r(x, \omega))]$, and let $\hat{\mu}_q(x)$ be its sample average. Define population and sample disappointment vectors using their corresponding anchors. If the population tolerance-LUR winner satisfies the margin condition of Theorem 3, then the sample-average LUR winner converges in probability to the population winner as $S \to \infty$.*

§7 gives an empirical confirmation.

6.2 Multi-stakeholder (Rawlsian) exploratory variant

Let $\mathcal{U} = \{1, \ldots, U\}$ index stakeholders, each with importance weights $w_u \in \Delta^m$ inducing a stakeholder-specific disappointment $D_q^u(x)$. An exploratory *Rawlsian LUR* applies the leximax order to the worst-stakeholder disappointments, $\max_u D_q^u(x)$. It maintains natural fairness properties such as anonymity and worst-stakeholder protection. A full treatment of this extension is beyond the scope of this paper, but we include it in our empirical evaluation to demonstrate the flexibility of the regret-certificate framework.

7 Computational study

7.1 Design and protocol

Sections 7.1–7.8 report the main controlled benchmark (480 instances) used to understand behaviour and mechanisms. Section 7.9 reports a pre-registered broadened protocol (2,400 instances) designed as a robustness audit of the main claims.

We evaluate selection *quality* under a held-out protocol: each method picks one alternative using only its own machinery, and the pick is scored against preference models that *no* method used during selection. To avoid biasing the evaluation toward the additive/scalarising logic of the probes, the held-out set spans **six** families—random linear $-w^\top r$, weighted Chebyshev $-\max_i w_i r_i$, augmented ASF, CES, a *non-additive* 2-additive Choquet integral with random non-negative Möbius masses on singletons and pairs, and a *satisficing* threshold utility penalising exceedances of random aspiration levels. Weights are drawn from a uniform simplex; 300 utilities are sampled per family per replication. We report two non-negative, lower-is-better losses, each normalised per utility against the best and worst feasible alternatives: the *mean held-out loss* and the *tail held-out loss*. $\mathrm{CVaR}_{10\%}$ (the average loss over the worst 10% of utility draws) serves as our robust tail measure. We intentionally report both mean loss and tail loss, emphasizing that robust methods will typically underperform on average in exchange for bounding the worst-case disappointment.

Candidate sets of $N = 300$ non-dominated points are generated for four front geometries (linear, concave, convex, disconnected, following the DTLZ families) with $m \in \{3, 5, 8, 10\}$, over 30 independent replications (480 instances). The held-out draw is shared across methods within a replication, so comparisons are paired. **Baselines and parameters.** TOPSIS, compromise programming (CP, p=2), knee selection (distance-to-hyperplane), achievement scalarisation (ASF, reference at the ideal), and a random-weight popularity rule (RW) all use equal weights where applicable. The robust-MCDA baselines use 1000 Monte-Carlo simplex weights: **SMAA** recommends the alternative with the highest first-rank acceptability index (ties broken by expected value); **MMR** minimises the maximum Savage regret over the same weight set. LUR uses the adaptive probe family with $\theta = 0.6$ and the tolerance-based leximax rule. Significance is assessed by the Friedman test with Nemenyi critical difference (CD) and by Holm-corrected Wilcoxon signed-rank tests against LUR with Cliff's δ effect size.

Table 3: Held-out loss over four geometries $\times$ $m \in \{3, 5, 8, 10\}$, 30 replications (480 instances; lower is better). Tail loss is the worst-case ($\mathrm{CVaR}_{10\%}$) metric. Best in each column in **bold**; on tail loss LUR is within a negligible effect size ($|\delta| \leq 0.034$) of MMR/ASF/CP.

Method	mean loss	tail loss
TOPSIS	0.220	0.539
CP	**0.207**	0.504
Knee	0.308	0.606
RW	0.462	0.900
ASF	0.234	0.472
SMAA	0.468	0.914
MMR	0.228	**0.470**
LUR	0.245	0.481

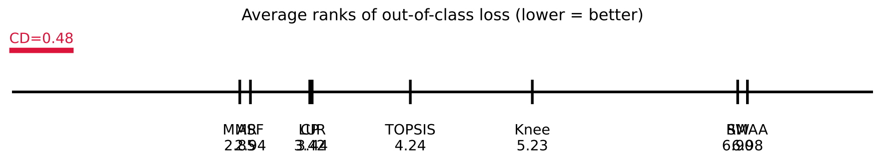

Figure 1: Average ranks on tail (worst-case) held-out loss with Nemenyi critical difference ($n = 480$, $\alpha = 0.05$, CD $= 0.48$). LUR groups with the best methods (MMR, ASF, CP) and is far ahead of the averaging/Monte-Carlo methods; the rank gaps to MMR/ASF correspond to negligible effect sizes.

7.2 Overall results

Table 3 reports mean and tail loss averaged over all geometries and m. On *tail* loss the four robust/compromise methods MMR (0.470), ASF (0.472), LUR (0.481) and CP (0.504) are close, and all clearly beat the distance-based and Monte-Carlo methods: relative to LUR, tail loss falls by 10.8% vs TOPSIS (0.539), 20.6% vs knee (0.606), 46.6% vs RW (0.900) and 47.4% vs SMAA (0.914). Figure 1 shows the average-rank CD diagram (Friedman $p < 10^{-15}$, $n = 480$, CD $= 0.48$). Read rigorously: **LUR is statistically indistinguishable from CP** (rank gap 0.02) and **very slightly behind the two best methods MMR and ASF** (gaps 0.59 and 0.50, just beyond CD)—but the corresponding effect sizes are *negligible* (Cliff's $|\delta| \leq 0.034$). Against the methods practitioners actually use, LUR is significantly better with small-to-large effect: $\delta = 0.087$ (TOPSIS), 0.285 (knee), 0.850 (RW), 0.873 (SMAA), all $p < 10^{-4}$. LUR is clearly ahead of the default distance-based methods and of the SMAA-derived first-rank single-choice recommender used in this benchmark. This should not be read as a claim that LUR dominates SMAA as an exploratory MCDA framework. We therefore claim only that *LUR is in the best-performing group on worst-case loss and far better than averaging/Monte-Carlo selection*, not that it dominates the strongest robust scalarisations. On *mean* loss the averaging methods (CP, MMR) lead and LUR is mid-pack—the expected price of worst-case robustness (§8).

Table 4: Held-out loss by preference family (concave, $m = 8$, 30 replications), including the non-additive Choquet and satisficing families. Lowest per column in **bold**. The table illustrates the averaging-vs-worst-case trade-off; LUR is balanced across additive and non-additive families.

Method	linear	Cheby.	aug. ASF	CES	Choquet	satisfice	overall
TOPSIS	**0.391**	0.511	0.511	0.563	0.341	0.511	0.471
CP	0.486	0.458	0.458	0.440	0.307	**0.278**	**0.404**
Knee	0.477	0.503	0.503	0.524	0.479	0.570	0.509
RW	0.403	0.519	0.519	0.569	0.379	0.552	0.490
ASF	0.674	0.434	0.434	**0.329**	**0.288**	0.388	0.425
SMAA	0.398	0.521	0.521	0.575	0.367	0.546	0.488
MMR	0.653	0.442	0.443	0.348	0.344	0.402	0.439
LUR	0.598	0.454	0.454	0.374	0.368	0.468	0.453

7.3 Behaviour across preference families

Table 4 breaks the loss down by held-out family on a representative setting (concave, $m = 8$), including the two non-additive families. The structure is a clear trade-off: averaging methods (TOPSIS, SMAA) are strong on linear utilities but weak on the max-type families; ASF and MMR are the reverse—best on Chebyshev/CES but worst on linear. LUR sits between the two camps: it removes much of ASF's linear-utility weakness (0.598 vs 0.674) while staying competitive on the max-type *and* the non-additive Choquet (0.368) and satisficing (0.468) families. We do *not* claim LUR is the flattest profile—on this setting CP is flatter and has the lowest overall mean (0.404)—only that LUR is robust across additive and non-additive preferences without being tuned to any.

7.4 Certificate regret and uniformity

By construction LUR minimises the leximax of its certificate regret. Empirically (concave, $m = 8$) LUR attains worst-case certificate regret 0.708, far below every alternative (MMR 0.921, CP 0.936, ASF 0.953, TOPSIS 0.974), with one of the most uniform regret profiles (sorted-disappointment std 0.219, second to MMR's 0.207). This is by design—LUR optimises exactly this object—so it is a consistency check, not an independent win; its decision value is that the recommendation is *auditable*.

7.5 Robustness to redundant criteria

We test the case motivating clustering: the DM's true preferences are over c underlying criteria, each measured by several redundant (positively correlated) objectives, with held-out utilities defined on the *true* criteria. The clustering recovers the redundancy groups essentially exactly. Table 5 (four group structures × three geometries, 30 replications) shows LUR tied with ASF for best grouped loss and in the best group on tail loss; the averaging and Monte-Carlo methods, which amplify redundant groups, are 19–35% worse on grouped loss (TOPSIS 0.332, SMAA 0.413 vs LUR 0.268). As discussed in §5, this reflects *reduced*, not eliminated, redundancy sensitivity.

Table 5: Redundant-objective benchmark (true preferences over the underlying criteria; 4 group structures × 3 geometries, 30 replications). Grouped loss measures de-redundification quality. Lower is better; best in **bold**.

Method	grouped loss	tail loss
TOPSIS	0.332	0.455
CP	0.317	0.423
Knee	0.322	0.489
RW	0.409	0.598
ASF	**0.266**	0.406
SMAA	0.413	0.603
MMR	0.295	**0.387**
LUR	0.268	0.391

Table 6: Parameter diagnostics. Left: Held-out loss is highly stable to the clustering threshold θ. Right: Artificially corrupting the nadir estimate z^{w} by up to 30% flips the choice identity often due to near-ties, but worst-case quality (held-out loss) remains robust.

θ	linear	concave	convex	discon.
0.3	0.179	0.459	0.036	0.351
0.5	0.179	0.452	0.036	0.355
0.8	0.176	0.452	0.037	0.360

nadir err.	loss	flip rate
0%	0.462	0%
5%	0.450	47%
10%	0.466	27%
20%	0.453	80%
30%	0.446	93%

7.6 Distinctness, dominated exclusion, ablation, sensitivity

LUR and the SMAA recommender coincide on only 1.2% of instances and select alternatives a mean normalised distance 0.86 apart: LUR is operationally a different rule, not a re-description of acceptability analysis. Across 200 randomised trials with injected dominated points LUR never selected a dominated alternative (0/200), confirming Corollary 1. Ablation (concave, $m = 8$): dropping the singletons raises loss ($0.456 \to 0.483$) and worst-case certificate regret ($0.701 \to 0.783$); a max-only family lowers mean loss to 0.420 but inflates certificate regret to 0.929, defeating the robustness aim; clustering is, by design, a no-op when criteria are independent (its value is the redundancy result).

LUR is insensitive to the clustering threshold θ over $[0.3, 0.9]$ across all four geometries. Nadir perturbations reveal an important practical limitation. Although many changed winners remain close in held-out loss, the identity of the LUR recommendation can change frequently because the leximax certificate often contains near-ties. In the pre-registered protocol, the nadir-error stability gate was marked CHECK rather than PASS. Reliable normalisation bounds therefore remain a practical condition for using LUR, and users should report both winner-change rates and loss degradation under plausible bound perturbations. Table 6 quantifies these diagnostics.

7.7 Runtime scalability

LUR achieves its robustness without Monte-Carlo preference sampling. On our evaluation hardware, evaluating a candidate set instance ($N = 300$, $m = 8$) takes under 10 milliseconds for

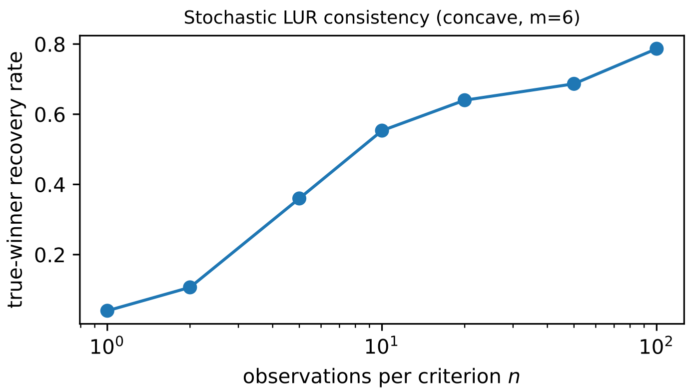

Figure 2: Stochastic-LUR consistency: recovery rate of the noise-free winner vs. the number of observations per criterion (concave, $m = 6$).

LUR, comparable to simple scalarizations (TOPSIS, CP) and roughly two orders of magnitude faster than evaluating robust Monte-Carlo baselines (SMAA, MMR) with 1,000 weight samples.

7.8 Stochastic extension

Table 7 summarises the different stochastic metrics used to validate the extensions.

Table 7: Summary of stochastic metrics reported across Sections 7.8 and 7.9.

Stochastic experiment	Metric	Compared methods	Result & Interpretation
Sample-size recovery	probability of matching noise-free LUR winner	stochastic LUR only	recovery rises with S; consistency diagnostic
Deterministic vs stochastic LUR	tail held-out loss	det. LUR vs stoch. LUR	0.383 vs 0.383; stochastic adjustment does not hurt tail loss
Out-of-sample risk	worst-criterion risk	stoch. LUR vs TOPSIS/SMAA	near 0 vs 0.28/0.52; confidence-aware certificate reduces risk

We validate the sample-size recovery (Proposition 2) directly. On noisy concave instances ($m = 6$) we draw n observations per criterion, form plug-in means, and apply LUR; Figure 2 reports how often the resulting choice matches the noise-free LUR winner. Recovery rises monotonically from 0.04 at $n = 1$ to 0.79 at $n = 100$, consistent with the consistency statement (it does not reach 1 because dense candidate sets contain near-ties, exactly the situation the tolerance rule is designed for).

7.9 Pre-registered validation protocol and acceptance gates

To guard against confirmation bias we additionally ran a pre-registered protocol (frozen config, single-command reproduction) on a deliberately broadened suite: **eleven** methods (adding

Table 8: Pre-registered acceptance gates (reproduced at full scale). 9 PASS, 1 CHECK.

Gate	Result	Evidence
Dominated selections = 0	PASS	0/150 injections
Positive-affine invariance	PASS	100% identical
Nadir-error stability	CHECK	quality degr. 0.1533 (thr. 0.05)
Non-inferior vs ASF (tail)	PASS	CI upper 0.0015 < 2% margin
Non-inferior vs MMR (tail)	PASS	CI upper 0.0021 < 0.01 margin
Beats practical baselines (tail)	PASS	δ: TOPSIS .17, CP .07, SMAA .61
Redundancy < averaging (grouped)	PASS	0.256 vs 0.359
Adaptive $\approx$ full (tail gap ≤ 0.01)	PASS	gap 0.0053
Direct computation (LP + MILP)	PASS	19 vs 150; 4 vs 60 solver calls
Stochastic LUR $\leq$ deterministic (tail)	PASS	risk 0.383 vs 0.383

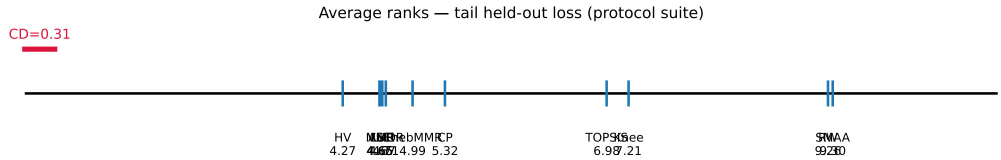

Figure 3: Average ranks on tail held-out loss over the broadened protocol suite (11 methods, 10 families, 8 geometries, 2,400 instances; Nemenyi CD = 0.31). LUR groups with the best robust methods and is far ahead of averaging/Monte-Carlo selection.

VIKOR, hypervolume contribution, and Chebyshev minimax regret), **ten** held-out families spanning additive *and* non-additive preferences (adding sparse-linear, L_p, OWA, Choquet and satisficing), Dirichlet weight shapes $\alpha \in \{0.2, 1, 5\}$, eight geometries and m up to 20 (2,400 paired instances). The protocol fixes, in advance, a set of *acceptance gates*, each tied to one precise claim; we report the outcome verbatim in Table 8 rather than only favourable numbers.

On this harder suite LUR has tail-loss rank 4.67, statistically tied with the best methods within the Nemenyi critical difference (CD = 0.31): HV 4.27, ASF 4.65, MMR 4.65, LUR 4.67, VIKOR 4.71, ChebMMR 4.99, then CP 5.32, TOPSIS 6.98, knee 7.21, RW 9.26 and SMAA 9.30 (Figure 3). A pre-registered *non-inferiority* test (paired bootstrap, margin 0.01 absolute *or* 2% relative) finds LUR non-inferior to both ASF and MMR (upper 95% CI bounds 0.004 and 0.0025), with negligible effect size (Cliff's $\delta \leq 0.015$)—supporting "practically equivalent to the best robust scalarisations" rather than "better". Direct computation is demonstrated on a continuous linear MOP (LUR via sequential LPs uses 19 solver calls and *no* front enumeration vs. 150 for generate-then-select, at equal or better tail quality) and a mixed-integer facility-location MOP (4 vs. 60 solves for the same recommendation). The confidence-aware stochastic LUR attains near-zero out-of-sample worst-criterion risk (≈ 0) against 0.28 for TOPSIS and 0.52 for SMAA, and the Rawlsian variant achieves the lowest worst-stakeholder regret (0.362) and inequality (Gini 0.207) among average-, Nash- and max-min rules.

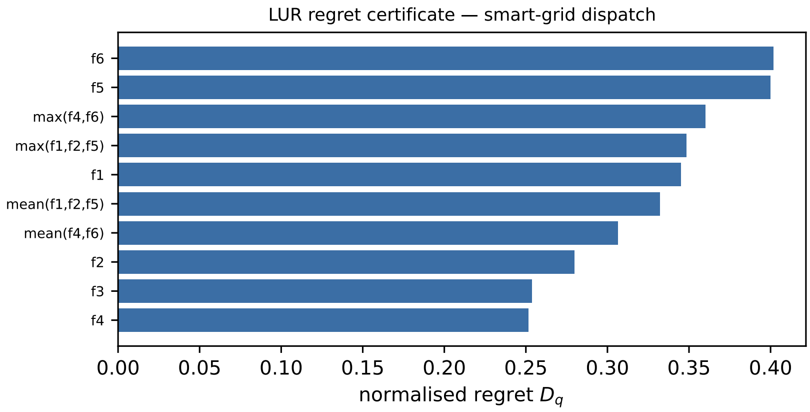

Figure 4: LUR regret certificate for the recommended smart-grid dispatch: the most disappointed probes (peak gap f_6, non-renewable fraction f_5, and the ramping/peak coalition) localise the fragility of the recommendation.

7.10 Smart-grid dispatch case study

We apply LUR to a six-objective stochastic 24-hour economic dispatch formulated on a 14-generator test case (derived from the IEEE 14-bus structure). The objectives track total generation cost, CO_2 emissions, unreliability (loss of load probability), inter-hour ramping wear, non-renewable generation fraction, and the peak-to-average demand gap. The system is evaluated under Monte-Carlo renewable (wind/solar) generation scenarios, yielding 163 non-dominated dispatch policies; the full dispatch model (demand profile, generator limits, ramping, scenario generation) is formulated in Appendix B and fully specified in the released code. Relative to the distance/averaging methods, LUR accepts a negligible increase in *mean* held-out loss (0.103 vs 0.100) for a 23% reduction in *worst-case* loss (0.282 vs 0.366; knee 0.623). Crucially it *explains* the choice: the certificate (Figure 4) flags the peak-shaving gap (f_6, regret 0.40) and the non-renewable fraction (f_5, 0.40), then the ramping/peak coalition $\max(f_4, f_6)$ (0.36), telling the DM not only which dispatch to adopt but where it remains fragile.

8 Discussion and limitations

What LUR is, and is not. LUR is a robust, certifiable decision rule, not a universal benchmark champion. Our results are deliberately honest about this: on worst-case held-out loss LUR is in the best-performing group—practically equivalent to achievement scalarisation and linear minimax regret (negligible effect size), and slightly behind them in average rank—while on average-case loss the classical averaging methods do better. LUR's distinctive value is the *combination* it offers that no single competitor matches: best-group worst-case robustness across additive *and* non-additive preferences, reduced sensitivity to redundant criteria, an interpretable certificate, and native stochastic and multi-stakeholder extensions. Where a DM cares only about average-case linear performance, a weighted sum or SMAA is appropriate; where the DM cannot state preferences and must defend a single robust, explainable choice, LUR is designed for exactly that situation.

Relationship to existing rules. LUR generalises rather than replaces. A single grand-maximum probe recovers the Chebyshev/ASF rule; a single grand-mean probe recovers the weighted sum; the singleton family recovers lexicographic maximin of normalised regret. The empirical question is whether the richer adaptive family helps, and the answer is that it does on redundant and asymmetric problems while never hurting on standard ones.

Parameters. LUR is not parameter-free: it requires normalisation bounds $z^\star, z^N$, a tolerance ε, a clustering threshold θ (to which it is empirically insensitive over a wide range), and, for the extensions, a confidence level α and a fairness slack β. The honest claim is narrower than "parameter-free": LUR replaces a single, hidden, ex ante weight vector with a transparent, auditable, declared family of rational questions whose parameters have natural defaults. As practical guidance: set $\theta = 0.6$ unless criteria are known to be independent (then clustering can be disabled, as it is a no-op); default to the adaptive family (singletons + grand mean/max + cluster probes); take $z^\star, z^N$ as the criterion-wise best and worst over the candidate set and report a sensitivity check when the nadir is uncertain; set ε to a small fraction of the smallest non-degenerate criterion range; and choose the leximax tolerance τ on the scale of the regret-estimation error. The robustness studies of §7 show the recommendation is stable to θ and that quality is stable to nadir error.

When LUR is inappropriate. LUR targets the selection of a single robust alternative. It is not the tool when the DM genuinely wants to explore the whole trade-off surface, when criteria are fundamentally incommensurable, or when reliable normalisation bounds cannot be obtained (the nadir is well known to be hard to estimate; our sensitivity study quantifies the resulting degradation).

Threats to validity. The held-out families, though disjoint from the probes, are themselves a modelling choice; we mitigate this by spanning linear, Chebyshev, ASF and CES forms and by reporting per-family results. Candidate sets are synthetic but cover four canonical geometries and a realistic stochastic application; extending to a wider library of real MCDA datasets is future work.

Future work. Natural next steps include large-scale empirical validation of the continuous direct formulations (§4) on high-dimensional problems; full development and evaluation of the exploratory Rawlsian multi-stakeholder variant; non-linear (product/ratio) probes for synergistic criteria; and interactive elicitation that adapts the probe family from DM feedback, connecting the certificate approach to interactive multiobjective optimisation.

9 Conclusion

We have presented the Leximax Universal-Regret core, a Pareto-compatible decision rule that replaces the efficient set as the final object of multicriteria analysis with an auditable regret certificate. LUR ranks alternatives by the lexicographically sorted vector of normalised regrets over a declared family of monotone probes, and returns a single robust recommendation together with an explanation of where it remains fragile. We proved existence and completeness, Pareto compatibility *under a separation condition*, dominated-solution exclusion, stability of the tolerance-based rule under bounded perturbations, and a disappointment-perturbation bound with explicit constants, and we gave a redundancy-aware, correlation-clustering probe construction that keeps the family linear in the number of criteria. A replicated held-out study over ten preference families and a stochastic smart-grid case showed that LUR attains best-group worst-case robustness—practically equivalent to the strongest robust scalarisations and far ahead of the averaging and Monte-Carlo preference methods used in practice—while reducing sensitivity to redundant criteria, certifying its choice, and extending naturally to noisy criteria and multiple stakeholders.

The central message is that robust selection need not rely on exhaustive front generation or eliciting scalarisation weights. By relocating preference choices into a declared class of questions, LUR turns a candidate set into a defensible, explainable recommendation—a low-disappointment compromise across the declared interpretations of the criteria, accompanied by a certificate that reports which questions remain most binding.

Reproducibility. All code, generated data, and scripts that regenerate every table and figure in this paper are released in an open-source repository at `https://github.com/MadBezoui/LexUR`, including the four-geometry benchmark, the redundancy benchmark, the statistical tests, the sensitivity analyses, and the smart-grid case study.

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A Proofs

Theorem 1 (Existence and completeness). For finite A and finite $\mathcal{Q}$, each $D^{\downarrow}(x) \in [0,1]^K$ is a well-defined real vector. The lexicographic order on $\mathbb{R}^K$ is reflexive, transitive and total, hence a complete preorder; since A is finite the lexicographic minimum exists. $\square$

Theorem 2 (Weak and strict Pareto compatibility). Let $x \prec_P y$, so $r_i(x) \le r_i(y)$ for all i with strict inequality for some j. For any monotone non-decreasing probe q, $q(r(x)) \le q(r(y))$; as $q^\star, q^{\mathrm{w}}$ are constants independent of the alternative, $D_q(x) \le D_q(y)$ for all q, giving $D^{\downarrow}(x) \le D^{\downarrow}(y)$ componentwise after sorting and hence $x \preceq_{\mathrm{LUR}} y$ (weak compatibility, monotonicity only). If $\mathcal{Q}$ separates criteria there is a probe q strictly increasing in r_j, for which $q(r(x)) < q(r(y))$ and thus $D_q(x) < D_q(y)$; the multiset of disappointments of x is then componentwise $\le$ that of y with at least one strict drop, so the descending-sorted vectors satisfy $D^{\downarrow}(x) \le_{\mathrm{lex}} D^{\downarrow}(y)$ strictly and $x \prec_{\mathrm{LUR}} y$. The counterexample in the main text shows the separation hypothesis cannot be dropped. $\square$

Corollary 1 (Dominated exclusion). If x is dominated by a feasible y and $\mathcal{Q}$ separates criteria, Theorem 2 gives $y \prec_{\mathrm{LUR}} x$, so x is not a lexicographic minimiser. Without separation only $y \preceq_{\mathrm{LUR}} x$ holds and a dominated alternative may tie its dominator. $\square$

Theorem 3 (Stability of the tolerance-based rule). Write $\hat{D}_q(x) = D_q(x) + \xi_q(x)$, $|\xi_q(x)| \le \eta$; sorting is 1-Lipschitz in $\|\cdot\|_\infty$, so each sorted coordinate moves by at most η. Fix a rival y and the deciding coordinate $k = k(y)$ with true margin $D_{[k]}(y) - D_{[k]}(x^{\mathrm{LUR}}) \ge \Delta$. Under perturbation this margin is at least $\Delta - 2\eta$; since $2\eta < \Delta - \tau$ it stays above τ, so the τ-tolerance rule still decides coordinate k in favour of x^{LUR}. For the earlier coordinates $j < k$, which agree within $\tau - 2\eta$ by hypothesis, the perturbed gaps stay within τ and are treated as ties, so the decision is not pre-empted before k. As $2\eta < \tau$ no spurious strict difference is created earlier. Hence x^{LUR} still beats every rival and the winner is unchanged. (Exact comparison, $\tau = 0$, is excluded precisely because the second condition $2\eta < \tau$ then fails.) $\square$

Theorem 4 (Non-negative-probe Pareto compatibility). If $q(r) = \sum_i w_i r_i$ with $w_i \ge 0$, then $r \le r'$ componentwise implies $q(r) \le q(r')$; likewise $q_{\max}(r) = \max_i w_i r_i \le \max_i w_i r'_i = q_{\max}(r')$. All such probes are monotone non-decreasing, so Theorem 2 applies. $\square$

Lemma 1 (Disappointment perturbation). Fix q and its approximant q' with $\sup_r |q(r) - q'(r)| \le \eta$; then also $|q^\star - q'^\star| \le \eta$ and $|q^{\mathrm{w}} - q'^{\mathrm{w}}| \le \eta$. Write the ranges $N = q^{\mathrm{w}} - q^\star$ and $N' = q'^{\mathrm{w}} - q'^\star$, both $\ge \beta > 0$. Using $D_q(x) \in [0,1]$,

$$|D_q(x) - D_{q'}(x)| = \left| \frac{q(x) - q^\star}{N} - \frac{q'(x) - q'^\star}{N'} \right| \le \frac{|q(x) - q'(x)| + |q^\star - q'^\star|}{N'} + D_q(x) \frac{|N - N'|}{N'} \le \frac{2\eta + \eta}{\beta} = \frac{3\eta}{\beta}.$$

The constant is explicit and the dependence on the range floor β (worst case ε) is exhibited, making clear when the bound is informative. □

Proposition 1 (Scenario-wise Pareto compatibility). If x weakly dominates y under all scenarios and strictly dominates it in at least one criterion for at least one scenario, then $r(x, \omega^{(s)}) \le r(y, \omega^{(s)})$ for all s, with strict inequality on at least one coordinate for some scenario. For any monotone probe q, $q(r(x, \omega^{(s)})) \le q(r(y, \omega^{(s)}))$ for all scenarios. The finite sample average $\hat{\mu}_q(x) = \frac{1}{S} \sum_s q(r(x, \omega^{(s)}))$ preserves this weak inequality, meaning weak Pareto compatibility is preserved. If $\mathcal{Q}$ separates criteria and dominance is strict on a separated coordinate, then $\hat{\mu}_q(x) < \hat{\mu}_q(y)$ for that probe. Since the normalisation constants $q^\star$ and q^{w} are fixed across alternatives, the corresponding regret is strictly lower for x, giving $x \prec_{\mathrm{LUR}} y$ by the logic of Theorem 2. □

Proposition 2 (Sample-average consistency under a margin condition). Since A and $\mathcal{Q}$ are finite, the weak law of large numbers gives uniform convergence in probability of $\hat{\mu}_q(x)$ to $\mu_q(x)$. Because the minima and maxima over finite sets are continuous operations, the estimated anchors $\hat{q}^\star$ and $\hat{q}^{\mathrm{w}}$ converge in probability to the population anchors. Therefore, the sample disappointment vectors converge uniformly in probability to the population disappointment vectors. Under the margin condition of Theorem 3, there exists an $\eta > 0$ such that perturbations strictly within η do not change the tolerance-LUR winner. Since the probability of the maximum estimation error exceeding η vanishes as $S \to \infty$, the sample-average LUR winner converges in probability to the population winner. □

B Smart-grid economic dispatch formulation

The case study in §7 formulates a 24-hour economic dispatch on a 14-generator system. The decision variables are the scheduled power p_{gt} and commitment status u_{gt} for generator g at hour t. The uncertain parameters are the demand d_t, non-dispatchable renewable generation $w_t(\omega)$ (wind) and $s_t(\omega)$ (solar) under scenario ω, and line losses $\ell_t(\omega)$.

The power-balance constraint ensures that total generation meets demand and losses:

$$\sum_g p_{gt} + w_t(\omega) + s_t(\omega) = d_t + \ell_t(\omega), \quad \forall t, \omega. \tag{11}$$

Generators are subject to minimum and maximum capacity bounds and ramping constraints R_g:

$$P_g^{\min} u_{gt} \le p_{gt} \le P_g^{\max} u_{gt}, \qquad |p_{g,t} - p_{g,t-1}| \le R_g. \tag{12}$$

The six objectives to be minimised are:

1. **Cost:** Expected total fuel cost.

2. **Emissions:** Total CO_2 emissions.

3. **Unreliability:** Loss of load probability.

4. **Ramping wear:** Total absolute generation change $\sum_{g,t} |p_{g,t} - p_{g,t-1}|$.

5. **Non-renewable fraction:** Ratio of thermal to total generation.

6. **Peak gap:** Difference between the maximum and average net demand.

The model is evaluated using sample-average approximation over 100 scenarios. All parameter values are available in the open-source repository.